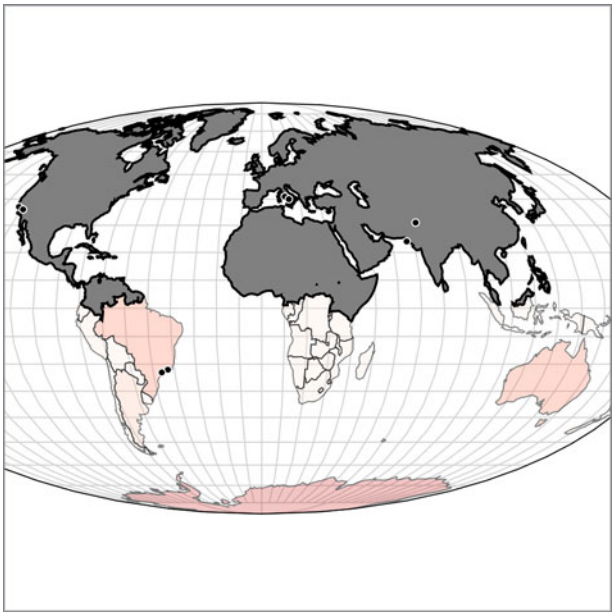


Figure 7.17 The results of merging based on the centroid of a feature. The feature in gray is a single merged feature made up of many separate polygons.

7.4.3 Merging

The TopoJSON library provides you with the capacity to create new features by merging existing features. You can create a new feature for "North America" by merging the countries in North America, or create "The United States in 1912" by merging the states that were part of the United States in 1912. Listing 7.17 shows the code to draw a map using our new TopoJSON data file and merge all the countries that have a center west of 0° longitude. The results, in figure 7.17, show that merging combines not only contiguous features but also separate features into a multipolygon.

Listing 7.17 Rendering and merging TopoJSON

```
queue()
  .defer(d3.json, "world.topojson")
  .defer(d3.csv, "cities.csv")
  .await(function(error, file1, file2) { createMap(file1, file2); });

function createMap(topoCountries, cities) {
  var countries =
     topojson.feature(topoCountries, topoCountries.objects.countries);
  var width = 500;
  var height = 500;
  var projection = d3.geo.mollweide()
     .scale(120)
     .translate([width / 2, height / 2])
     .center([20,0]);

  var geoPath = d3.geo.path().projection(projection);

  var featureSize =
     d3.extent(countries.features, function(d) {return geoPath.area(d)});
```